Exercises 3.1

1) Let A be an input to insertion sort of size n. Suppose the $\lfloor n/3 \rfloor$ largest values occupy the first $\lfloor n/3 \rfloor$ positions $A[1 : \lfloor n/3 \rfloor]$. Once the array is sorted, each of these $\lfloor n/3 \rfloor$ values end up in the last $\lfloor n/3 \rfloor$ positions by passing through at least $\lfloor n/3 \rfloor$ positions one position at a time ~~vo~~ via at least $\lfloor n/3 \rfloor$ executions of line 6. Hence the time taken by insertion sort in the worst case is at least $(\lfloor n/3 \rfloor)(\lfloor n/3 \rfloor)$ $\geq (n/3 - 1)(n/3 - 1) = n^2/9 - 2n/3 + 1 = \Omega(n^2)$.

2) Selection sort has two nested for loops, where the outerloop ~~always~~ iterates $n-1$ times and for each iteration of the outerloop, the innerloop iterates $n-i$ times (where i is the loop ~~variable~~ index of the outer~~t~~ loop). The body of the inner loop takes constant time per iteration. ~~There~~ Hence, there are always number of inner iterations

$$\sum_{i=1}^{n-1} (n-i) = (n-1) + (n-2) + \cdots + 2 + 1$$

$$= \frac{n(n-1)}{2} = \frac{n^2 - n}{2}$$

$$= O(n^2)$$

Since $(n^2-n)/2$ iterations of the inner loop is true for every case of SELECTION SORT, it is also $\Omega(n^2)$. Hence it is $\Theta(n^2)$

3) Let A be an input to insertion sort of size n, where the largest αn values start in the first αn positions. ~~In the final sorted array for some~~ where $0 < \alpha < 1$.

In the final sorted array, these αn values have to pass through the middle $(1-2\alpha)n$ array positions. To ensure these middle positions exist:-

$$(1-2\alpha)n > 0$$
$$1-2\alpha > 0$$
$$1 > 2\alpha$$
$$\alpha < 1/2$$

The total number of elements passing through these middle positions is at least $\alpha n((1-2\alpha)n) = \alpha(1-2\alpha)n^2$ which gives a lower bound $\Omega(n^2)$.

To maximise the number of times the αn largest values pass through the middle:-

$$f(n) = \alpha(1-2\alpha)$$
$$f(n) = \alpha - 2\alpha^2$$
$$f'(n) = 1-4a$$
$$\cancel{1-4a=}$$

Largest value of $\alpha(1-2\alpha)n^2$ when ~~α~~

$$1-4a = 0$$
$$4a = 1$$
$$a = 1/4$$